



CCNP IN PROGRESS -_-

Soho Configuration For a PA220 Pan OS (10.2.6)

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Cybersecurity Lab Report 2: Soho Initial configuration of PA-220

Purpose

Initial configuration to grant normal status for use on a home network connected to the internet.

Background Information on lab concepts

Concept: Understand the GUI of palo alto

Concept: Configuration by GUI

Background:

Features of the PA220

- Machine learning enhanced protection.
Through machine learning and algorithmic protection we can prevent new spoofing and phishing attacks without need to update the OS version allowing for greater security within your network
- Full layer 7 inspection with categorization of apps ports 24/7
- Adaptive policy capable to conform to user activity, capable of grouping outside of Ip addresses and other common grouping methods.
- Encrypts traffic inbound and outbound on TLS 1.3 and HTTP/2
- Single pass packet processing unique to palo alto.
Policy lookup decoding and threat content is scanned in one pass to reduce consumption and overhead with only one device.

Compared to cisco firewalls that are expensive and hard to use even for experts. Palo alto aims to gain popularity for ease of use makes waves in the early stages of learning for beginning cyber experts.

Lab Summary

Use (Show system info) to get the Ip.

Use Ip to set your own Ip on same network.

Once done you can type the ip into the search bar of your browser in the format HTTPS.

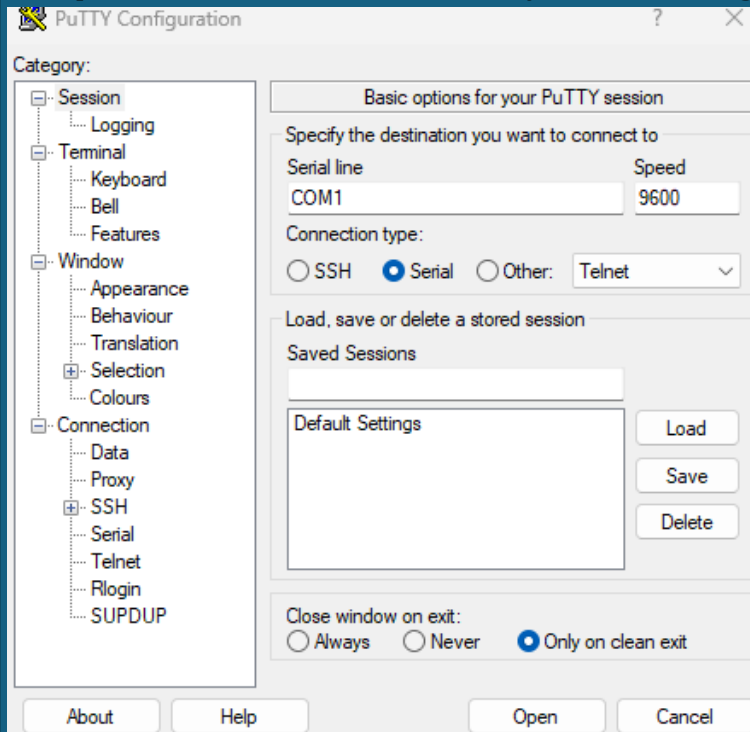
Configure Firewall for security

Configure for DHCP Incoming and outgoing

Full Guide:

Assuming that you have access to the firewall from the last lab and you factory reset it. You should use Admin user and Admin Password for default log on or use your pre-designated sign in for the next steps.

Gain primitive access to the Firewall by console through Putty. (console program)



It should look something like this after you press open on your open com. If its blank then press enter and wait ~ one minute. If not than check your console connection and verify the cord integrity.

```
PA-220 login: admin
Password:
Last login: Thu Sep 19 11:46:58 on ttyS0

Number of failed attempts since last successful login: 0

admin@PA-220>
admin@PA-220> █
```

Using the Show System Info Command in the console line you can find your default or currently configured IP along with other info like subnet class and netmask,

```
admin@PA-220> show system info

hostname: PA-220
ip-address: 192.168.1.1
public-ip-address: unknown
netmask: 255.255.255.0
default-gateway: 192.168.1.254
ip-assignment: static
ipv6-address: unknown
ipv6-link-local-address: fe80::c624:56ff:fe56:7900/64
ipv6-default-gateway:
mac-address: c4:24:56:56:79:00
time: Thu Sep 19 11:51:09 2024
uptime: 6 days, 5:14:51
family: 220
model: PA-220
serial: 012801110280
cloud-mode: non-cloud
sw-version: 10.2.6
global-protect-client-package-version: 0.0.0
device-dictionary-version: 1-211
device-dictionary-release-date:
app-version: 8614-7547
app-release-date:
lines 1-23
```

Once you find what network you need to be on to access the Web Gui, go find your “Ethernet Settings” in your windows search.

This will take you to your windows 11 Settings page or in windows 10 a different screen where you change IP’s through “properties”.

In that case select

I have Windows 11, so this is what mine looks like.

Network & internet > Ethernet



Unidentified network
No internet



Authentication settings

Edit

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Off ☐

[Set a data limit to help control data usage on this network](#)

IP assignment:

Manual

IPv4 address:

192.168.1.2

IPv4 mask:

255.255.255.0

Edit

DNS server assignment:

Automatic (DHCP)

Edit

Link speed (Receive/Transmit):

1000/1000 (Mbps)

Copy

Link-local IPv6 address:

fe80::a8db:d012:4171:e661%16

IPv6 DNS servers:

fec0:0:0:ffff::1%1 (Unencrypted)

fec0:0:0:ffff::2%1 (Unencrypted)

fec0:0:0:ffff::3%1 (Unencrypted)

IPv4 address:

192.168.1.2

Manufacturer:

Intel

Description:

Intel(R) Ethernet Connection (17) I219-LM

Driver version:

12.19.2.57

Physical address (MAC):

04-D9-C8-BA-25-EC



Ethernet 2
Not connected



To change a configuration manually (we haven't set auto up yet) set it to a useable Ip like 192.168.1.2 using the edit button.

Network & internet > Ethernet



Unidentified network
No internet



Authentication settings

Edit

Metered connection

Some apps might work differently to reduce data usage when you're connected to this network

Off ☐

[Set a data limit to help control data usage on this network](#)

IP assignment: Manual
IPv4 address: 192.168.1.2
IPv4 mask: 255.255.255.0

Edit

DNS server assignment: Automatic (DHCP)

Edit

Link speed (Receive/Transmit): 1000/1000 (Mbps)
Link-local IPv6 address: fe80::a8db:d012:4171:e661%16
IPv6 DNS servers: fec0:0:0:ffff::1%1 (Unencrypted)
fec0:0:0:ffff::2%1 (Unencrypted)
fec0:0:0:ffff::3%1 (Unencrypted)

Copy

IPv4 address: 192.168.1.2
Manufacturer: Intel
Description: Intel(R) Ethernet Connection (17) I219-LM
Driver version: 12.19.2.57
Physical address (MAC): 04-D9-C8-BA-25-EC



Ethernet 2
Not connected



Clicking the edit button will result in this menu, click on the word DHCP, and set it to manual or if you are already set to manual skip this step.

Edit IP settings

Automatic (DHCP)

Manual

Save

Cancel

When you select manual, it will open a plethora of options. All you need to do though is set both the IP address and the Subnet mask to the ones shown in your earlier, Show systems command. After its changed it should look like this. Leave the other settings as blank or off.

Edit IP settings

Manual

IPv4

☒ On

IP address

192.168.1.2

Subnet mask

255.255.255.0

Gateway

Preferred DNS

DNS over HTTPS

Off

Alternate DNS

Save Cancel

Clicking save will confirm the changes, if it doesn't just config it again and try to save it again. Technology isn't perfect either.

Now that you can access the same network as the firewall, make sure that you are ethernet connected with a normal ethernet cable into the management port of the firewall, the other ports wont work for GUI access unless you set that up.

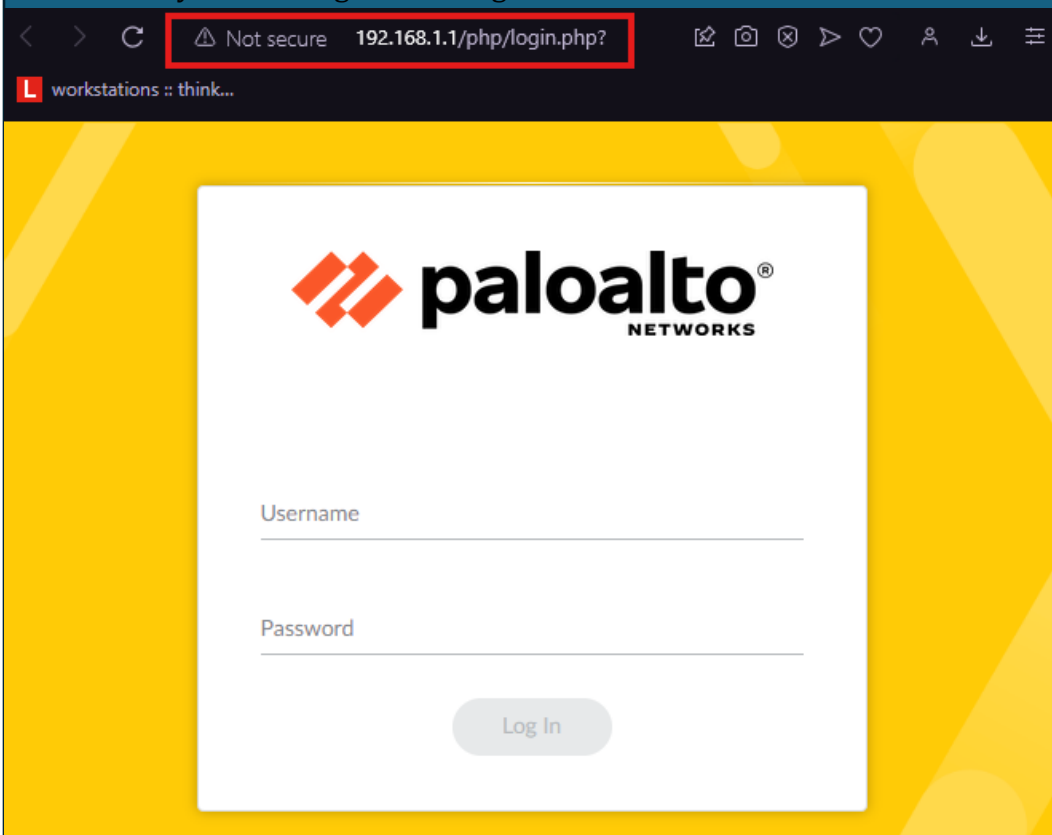


Ethernet connection from MGT to Ethernet on PC

Now using whatever browser you have, google, Firefox, etc.

Enter [HTTPS://192.168.1.1](https://192.168.1.1) into your search bar.

If successful you should get a message like this.



If you get a, "this site is trying to steal your identity" just click read or more info and then proceed to go to the website.

Login to palo alto with the username Admin and whatever password you set, or if you didn't set one, use admin. (default setting)



Create Security Zones

Go to: Network > Zones and click Add.

Create 3 zones:

- 1. Untrust-L3, Type Layer3
- 2. Trust-L3, Type Layer3
- 3. Trust-L2, Type Layer2

☐	trust	virtual-wire	
☐	Trust-L2	layer2	ethernet1/2
			ethernet1/3
			ethernet1/4
☐	Trust-L3	layer3	vlan
☐	untrust	virtual-wire	
☐	Untrust-L3	layer3	ethernet1/1

Here's the steps to set up trusted and untrusted zones.

192.168.1.1/?#dashboard:vsys1

workstations :: think...

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS **NETWORK** DEVICE

Layout 3 Columns Widgets Last updated 12:24:59

General Information

Device Name	PA-220
MGT IP Address	192.168.1.1
MGT Netmask	255.255.255.0
MGT Default Gateway	192.168.1.254
MGT IPv6 Address	unknown
MGT IPv6 Link Local Address	fe80::c24:56ff:fe56:7900/64
MGT IPv6 Default Gateway	
MGT MAC Address	c4:24:56:56:79:00
Model	PA-220
Serial #	012801110280
Software Version	10.2.6
GlobalProtect Agent	0.0.0
Application Version	8614-7547
Device Dictionary Version	1-211
URL Filtering Version	0000.00.00.000
GlobalProtect Clientless VPN Version	0
Time	Fri Sep 20 12:05:28 2024
Uptime	7 days, 5:29:10
Advanced Routing	off
Plugin DLP	dlp-3.0.5
Device Certificate Status	None

Logged In Admins

Admin	From	Client	Session Start	Idle For
admin	192.168.1.2		09/20/2024 12:05:06	00:00:12s

Data Logs

No data available.

System Logs

Description	Time
User admin logged in via Web from 192.168.1.2 using https	09/20 12:05:05
authenticated for user 'admin'. From: 192.168.1.2.	09/20 12:05:05
Port MGT: Up 1Gb/s Full duplex	09/20 12:02:59
Port MGT: Down 1Gb/s Full duplex	09/20 12:02:57
Port MGT: Up 10Mb/s Full duplex	09/20 12:02:55
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/20 11:57:02
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/20 11:41:12
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/20 11:26:53
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/20 11:11:30

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS **NETWORK** DEVICE

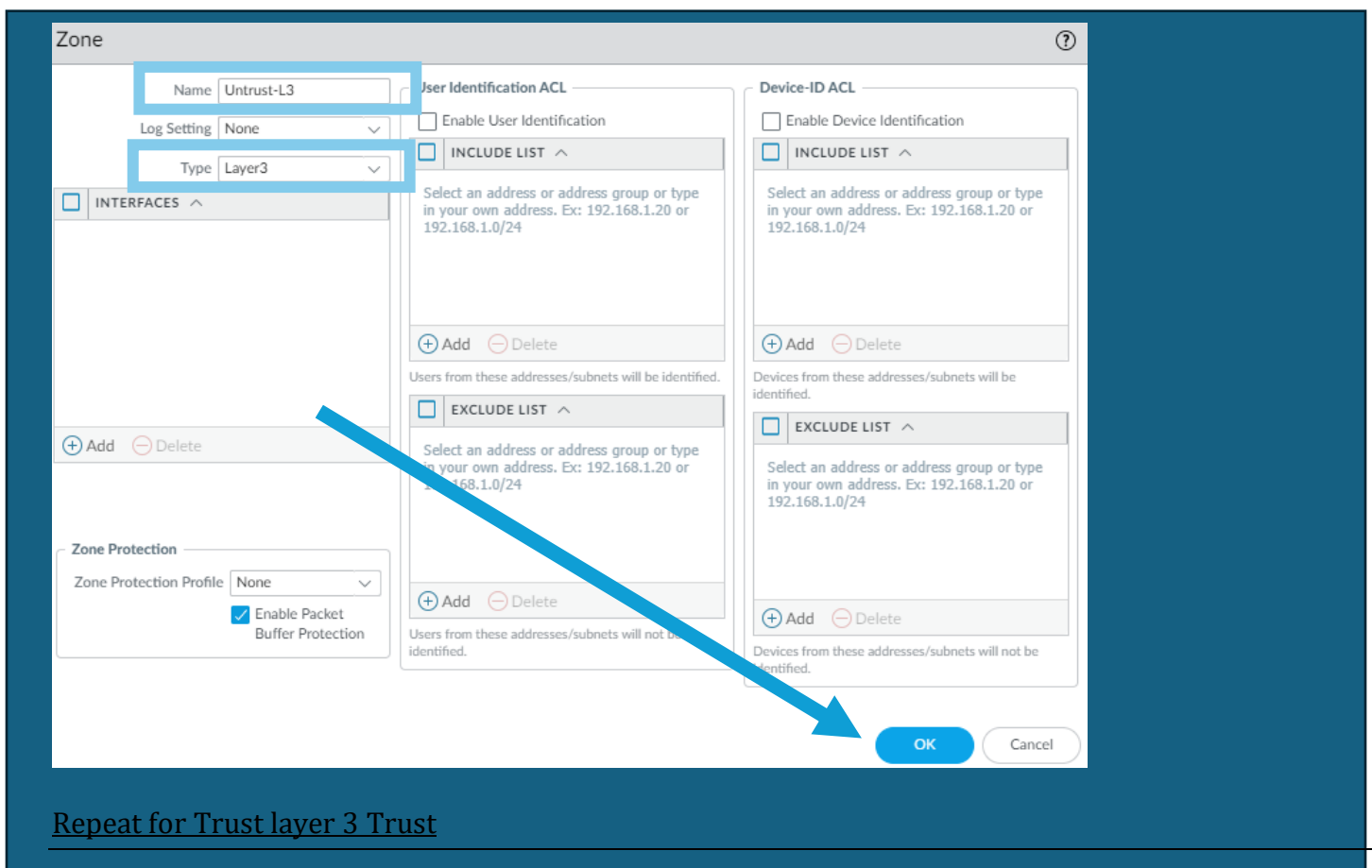
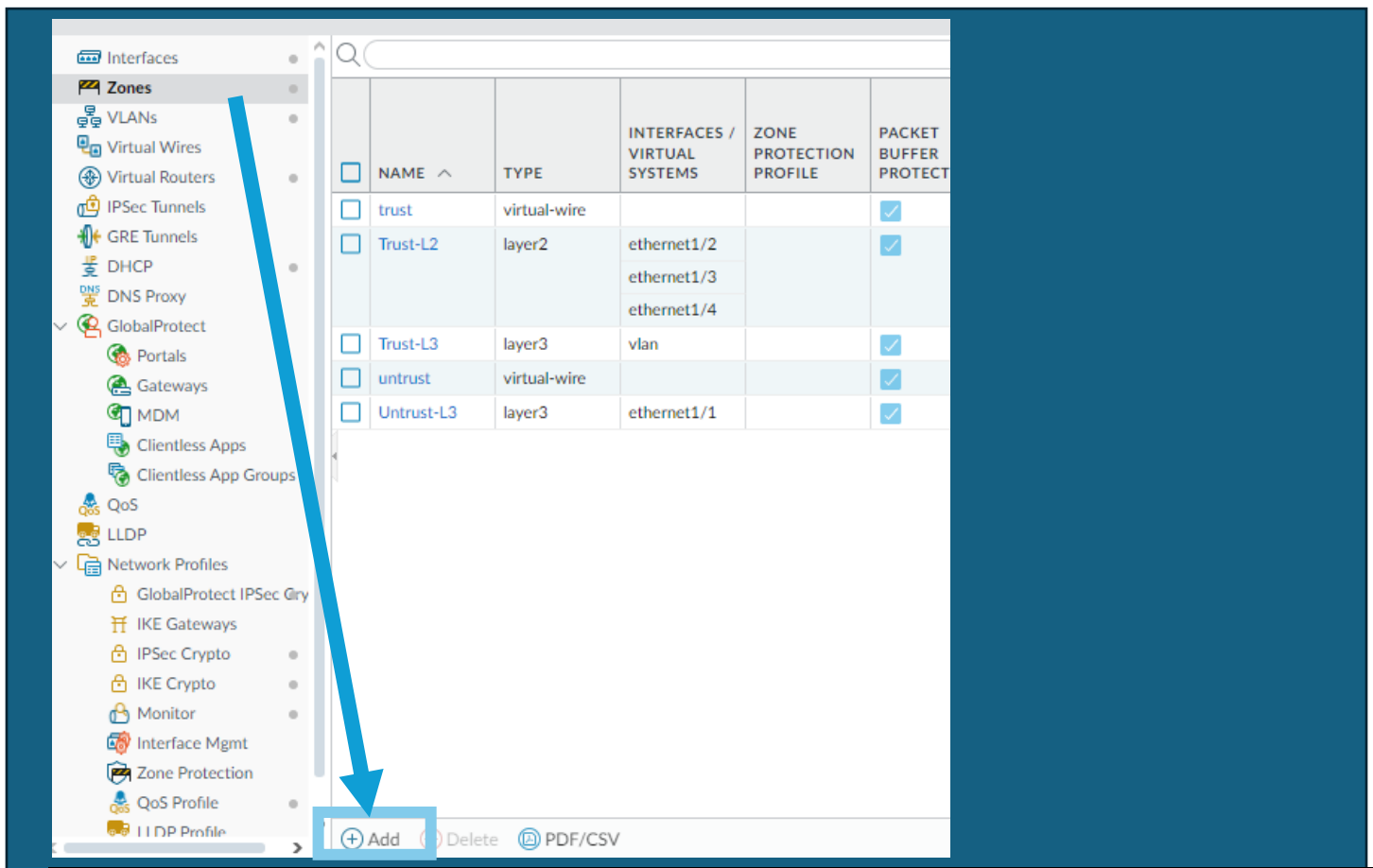
Interfaces

NAME	INTERFACE1	INTERFACE2	TAG ALLOWED
------	------------	------------	-------------

Virtual Wires

- Virtual Routers
- IPSec Tunnels
- GRE Tunnels
- DHCP
- DNS Proxy
- GlobalProtect
 - Portals
 - Gateways
 - MDM
 - Clientless Apps
 - Clientless App Groups
- QoS
- LLDP
- Network Profiles
 - GlobalProtect IPSec Gwy
 - IKE Gateways
 - IPSec Crypto
 - IKE Crypto
 - Monitor
 - Interface Mgmt
 - Zone Protection
 - QoS Profile
 - LLDP Profile

+ Add - Delete PDF/CSV



Repeat for Trust layer 3 Trust

For configuring Layer 2 Trust do this:

Name

Log Setting

Type

You should end with a list like this.

☐	trust	virtual-wire			☒		☐	any	none	☐	any	none
☒	Trust-L2	layer2	ethernet1/2		☒		☐	any	none	☐	any	none
			ethernet1/3									
			ethernet1/4									
☐	Trust-L3	layer3	vlan		☒		☐	any	none	☐	any	none
☐	untrust	virtual-wire			☒		☐	any	none	☐	any	none
☐	Untrust-L3	layer3	ethernet1/1		☒		☐	any	none	☐	any	none

This is where that internet connection to your wall comes in, I recommend using ethernet 1/1 like me.

Start by going to the Network tab up top. (same tab as previous step)

And navigate to Interfaces.

Interfaces | Ethernet | VLAN | Loopback | Tunnel | SD-WAN

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE
ethernet1/1	Layer3			Dynamic-DHCP Client	default	Untagged	none	Untrust-L3	

In Ethernet 1/1

Ethernet Interface

Interface Name

Comment

Interface Type

Netflow Profile

Config | IPv4 | IPv6 | SD-WAN | Advanced

Assign Interface To

Virtual Router

Security Zone

OK Cancel

Then enter Ipv4

You will find a little button that says DHCP Client

Click that and save.

Inside of VLANs:

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICE Commit

Interfaces Zones VLANs Virtual Wires Virtual Routers IPSec Tunnels GRE Tunnels DHCP DNS Proxy GlobalProtect

NAME	INTERFACES	VLAN INTERFACE
☒ Vlans8	ethernet1/2 ethernet1/3 ethernet1/4	vlan

Use the Add to create a VLAN for home network use.

VLAN ?

Name

VLAN Interface None ▾

Static MAC Configuration

☐ INTERFACES ^	MAC ADDRESS	INTERFACE
+ Add - Delete		

OK Cancel

In the name box you should Name it whatever is convenient. I named mine, Vlan8.

Select “vlan” as your VLAN Interface. (default vlan)

VLAN ?

Name Vlans8

VLAN Interface vlan ▾

Static MAC Configuration

☐ INTERFACES ^	MAC ADDRESS	INTERFACE
☐ ethernet1/2		
☐ ethernet1/3		
☐ ethernet1/4		
+ Add - Delete		

OK Cancel

Then confirm by pressing OK.

Making A DHCP Sever inside of the firewall.

Navigation: Go to networks -> DHCP -> DHCP Server (at the top inside the window)

The screenshot shows the PA-220 firewall configuration interface. The top navigation bar includes DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, and NETWORK. The left sidebar lists various network configuration options: Interfaces, Zones, VLANs, Virtual Wires, Virtual Routers, IPsec Tunnels, GRE Tunnels, DHCP (selected), and DNS Proxy. The main content area is titled 'DHCP Server | DHCP Relay'. Below the title is a search bar and a table with the following columns: INTERFACE, MODE, and PROBE IP. The table contains one entry: 'vlan' with 'enabled' mode and an unchecked 'PROBE IP' checkbox.

INTERFACE	MODE	PROBE IP
vlan	enabled	☐

Down at the Bottom, Click Add.

The screenshot shows the DHCP Server configuration form. The 'Interface' dropdown is set to 'vlan' and the 'Mode' dropdown is set to 'enabled'. The 'Lease' tab is selected, showing options for 'Ping IP when allocating new IP' (unchecked), 'Lease' type (Unlimited or Timeout, with Timeout selected), and lease duration (0 Days, 1 Hours, 0 Minutes).

Interface:

Mode:

Lease | Options

☐ Ping IP when allocating new IP

Lease: ☐ Unlimited ☒ Timeout

Days Hours Minutes

Select vlan as your interface.

Use enabled instead of auto to force it to run.

And lastly Select timeout and set it to an hour so you don get a buildup of devices.

Now click over to options and copy these for initial config.

The screenshot shows the DHCP Server configuration form, specifically the 'Options' tab. It displays various configuration fields for inheritance source, gateway, subnet mask, DNS, WINS, NIS, NTP, POP3, SMTP, and DNS suffix.

Inheritance Source:

[Check inheritance source status](#)

Gateway:

Subnet Mask:

Primary DNS:

Secondary DNS:

Primary WINS:

Secondary WINS:

Primary NIS:

Secondary NIS:

Primary NTP:

Secondary NTP:

POP3 Server:

SMTP Server:

DNS Suffix:

Configure L2 Ports and Vlan Object

Navigation: Network -> Interfaces -> Ethernet (2,3,4)

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS **NETWORK**

Interfaces • Zones • VLANs • Virtual Wires • Virtual Routers • IPSec Tunnels • GRE Tunnels • DHCP • DNS Proxy • GlobalProtect • Portals • Gateways • MDM • Clientless Apps • Clientless App Groups • QoS • LLDP • Network Profiles

Ethernet | VLAN | Loopback | Tunnel | SD-WAN

Q

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER
 ethernet1/1	Layer3			Dynamic-DHCP Client	default
 ethernet1/2	Layer2			none	none
 ethernet1/3	Layer2			none	none
 ethernet1/4	Layer2			none	none
 ethernet1/5				none	none
 ethernet1/6				none	none
 ethernet1/7				none	none
 ethernet1/8				none	none

And by clicking on Ethernet 1/2 you can enter its config.

Ethernet Interface ?

Interface Name

Comment

Interface Type

Netflow Profile

Config | Advanced

Assign Interface To

VLAN

Security Zone

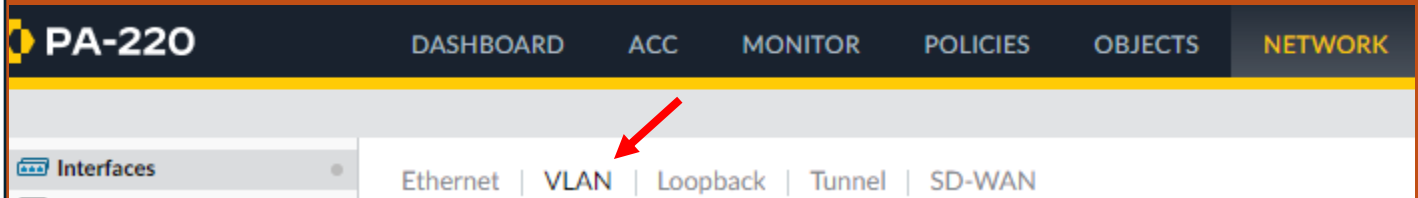
OK Cancel

Use your configured vlan and select interface type as Layer 2.

Any ports you intend to actively use should have this process completed before use, otherwise they will not function properly.

Now to configure your VLAN interface

Navigation: Network -> Interfaces -> VLAN



Set The Vlan to What you named it before, again. And use the default Virtual Router, then set this to L3 Trust as this is going to be an internal Vlan. (trust inside traffic)

The screenshot shows the 'VLAN Interface' configuration window. The 'Interface Name' is 'vlan'. The 'Config' tab is selected. Under 'Assign Interface To', the 'VLAN' dropdown is set to 'Vlans8', 'Virtual Router' is set to 'default', and 'Security Zone' is set to 'Trust-L3'. The 'OK' button is highlighted.

Navigate to the Ipv4 Tab and enter this Ip: 192.168.1.254

The screenshot shows the 'VLAN Interface' configuration window with the 'IPv4' tab selected. The 'Type' is set to 'Static'. The 'IP' section is expanded, showing the IP address '192.168.1.254/24' entered in the 'New Address' field.

Now you have a configured Vlan with a IP that we will use later... on to security profiles.

Define a Security Profile Group

Navigation: Objects -> Security Profile Groups

PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS

Addresses
Address Groups
Regions
Dynamic User Groups
Applications
Application Groups
Application Filters
Services
Service Groups
Tags
Devices
GlobalProtect
HIP Objects
HIP Profiles
External Dynamic Lists
Custom Objects
Data Patterns
Spyware
Vulnerability
URL Category
Security Profiles
Antivirus
Anti-Spyware
Vulnerability Protection
URL Filtering
File Blocking
WildFire Analysis
Data Filtering
DoS Protection
Security Profile Groups

Search

☐	NAME	LOCATION	ANTIVIRUS PROFILE	ANTI-SPYWARE PROFILE
☐	Internet		default	strict

For creating a secure network and making use of the firewalls capabilities we will be using this config.

Security Profile Group ?

Name

Antivirus Profile

Anti-Spyware Profile

Vulnerability Protection Profile

URL Filtering Profile

File Blocking Profile

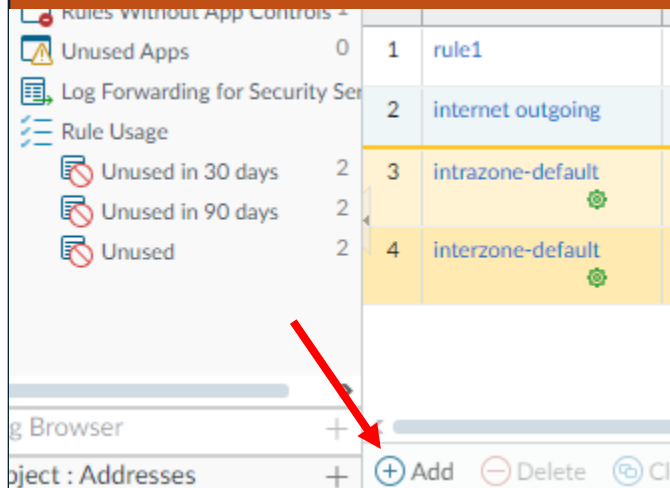
Data Filtering Profile

WildFire Analysis Profile

OK Cancel

Configuring Outbound Internet Security Policies

Navigation: Policies -> Security then click add



Now add a Name and Description

General	Source	Destination	Application
Name: internet outgoing			
Rule Type: universal (default)			
Description: All traffic to the internet			
Tags:			
Group Rules By Tag: None			
Audit Comment:			

Set your source zone to Trust-L3

General	Source	Destination	Application
☐ Any			
☒ SOURCE ZONE ^			
☐ Trust-L3			

And your Destination zone to Untrust-L3

General | Source | **Destination**

select

☐ DESTINATION ZONE ^

☐ Untrust-L3

And then set these Settings To “Actions”

General | Source | Destination | Application | Service/URL Category | **Actions** | Usage

Action Setting

Action

Allow

☐ Send ICMP Unreachable

Profile Setting

Profile Type

Group

Group Profile

Internet

Log Setting

☐ Log at Session Start

☒ Log at Session End

Log Forwarding

None

Other Settings

Schedule

None

QoS Marking

None

☐ Disable Server Response Inspection

And you are done with Security Policies, Lets move onto Internet NAT policies.

Navigation: Policies -> NAT

Lets set up an internet Outgoing Nat Policy

PA-220

DASHBOARDACCMONITOR**POLICIES**OBJECTSNETWORKDEVICE

Security

NAT

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

Authentication

DoS Protection

SD-WAN

			Original		
	NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE
1	Internet Outgoing	none	Trust-L3	Untrust-L3	ethernet1/1

In General we can name the Policy

Add a Description and select Ipv4 if not already selected.

NAT Policy Rule



General | Original Packet | Translated Packet

Name	Internet Outgoing
Description	
Tags	
Group Rules By Tag	None
NAT Type	ipv4
Audit Comment	

Add Trust L3 to your Original packet area (inside your network)

General Original Packet Translated Packet	
☐ Any	Destination Zone
☒ SOURCE ZONE ^	Untrust-L3
☐ Trust-L3	
	Destination Interface
	ethernet1/1
	Service
	any

And configure your Translated Packet Section to look like this.

General Original Packet Translated Packet	
Source Address Translation	Destination Address Translation
Translation Type: Dynamic IP And Port	Translation Type: None
Address Type: Interface Address	
Interface: ethernet1/1	
IP Address: None	

Press OK to save this change and now we are going to configure MGMT IP

Configure the MGMT IP

Navigation: Device -> Setup -> Interface Settings

The screenshot shows the top navigation bar of the PA-220 interface with tabs: DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and **DEVICE** (highlighted with a red box). Below the navigation bar, the left sidebar contains a 'Setup' icon (highlighted with a red box) and a list of setup options: High Availability, Config Audit, and Password Profiles. The main content area shows a breadcrumb trail: Management | Operations | Services | **Interfaces** (highlighted with a red box). Below the breadcrumb, a table lists the interfaces. A red arrow points to the 'Management' interface row.

INTERFACE NAME	ENABLED	SPEED
Management	☒	auto-negotia

This is where you can change the management IP and subnet and configure it to work for you. The original 192.168.1.1 works in my situation so that is what I will be continuing with.

Management Interface Settings ?

IP Type ☒ Static ☐ DHCP Client

IP Address

Netmask

Default Gateway

IPv6 Address/Prefix Length

Default IPv6 Gateway

Speed

MTU

Administrative Management Services

☐ HTTP ☒ HTTPS

☐ Telnet ☒ SSH

Network Services

☐ HTTP OCSP ☒ Ping

☐ SNMP ☐ User-ID

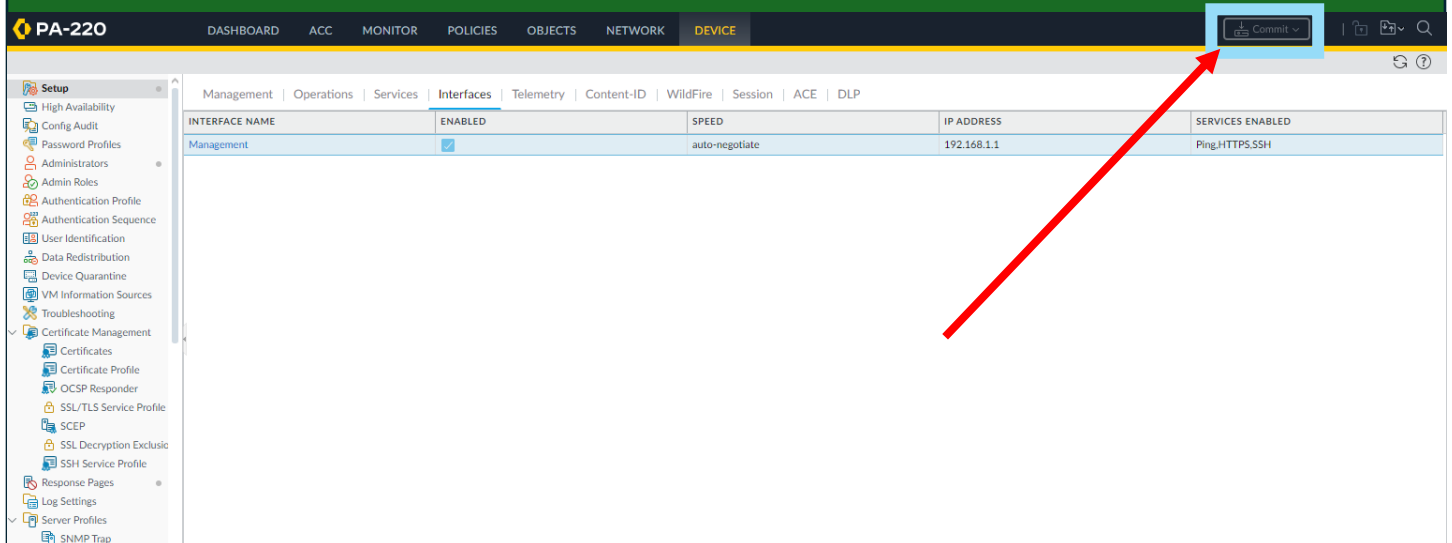
☐ User-ID Syslog Listener-SSL ☐ User-ID Syslog Listener-UDP

☐	PERMITTED IP ADDRESSES	DESCRIPTION
--------------------------	------------------------	-------------

COMMIT CHANGES!!!

Without committing changes you won't have your configuration updated and your work will delete upon restart of your firewall.

This is the final saving steps.



PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK DEVICES

Commit

Management | Operations | Services | Interfaces | Telemetry | Content-ID | WildFire | Session | ACE | DLP

INTERFACE NAME	ENABLED	SPEED	IP ADDRESS	SERVICES ENABLED
Management	☒	auto-negotiate	192.168.1.1	Ping,HTTPS,SSH

Now commit your changes.

Commit

Doing a commit will overwrite the running configuration with the commit scope.

☒ Commit All Changes ☐ Commit Changes Made By:(1) admin

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS
device-and-network	Device and Network Configuration			

[Preview Changes](#) [Change Summary](#) [Validate Commit](#)

Note: This shows all the changes in login admin's accessible domain.

Description

[Commit](#) [Cancel](#)

If you have Virtual Wire issues, try deleting the default virtual wire. It fixed the problem in most cases.

Good luck with your continued Firewall Configuration!

Cybersecurity Sign off Sheet

Notes/Issues and Stamp area below.

Lab Sotto Firewall



Name Careen

Date 2/23/2022



